

of the remaining diverse outputs, highlighting a complex trade-off between alignment objectives and output diversity.

Our results suggest that alignment algorithms should be designed with output diversity in mind. Future work should investigate how to strike a better balance in alignment algorithms to achieve both safety and helpfulness objectives while maintaining the rich diversity that makes language models useful for creative tasks.

5 Discussion and Future Work

Our findings suggest that language models lack *distributional* diversity; given a fixed prompt, their outputs are heavily skewed toward certain responses over other equally valid ones. Users can to some extent elicit novel generations by modifying their prompt (for example, adding more detailed instructions) or else iteratively asking “give me a different answer” in a multi-turn fashion. However, these workarounds are cumbersome and don’t address the fundamental issue that modern language models output overly low-entropy distributions which limit their generative capabilities. Methods that directly improve distributional diversity remain valuable as they will ensure diverse outputs across all generation settings, regardless of prompting strategy.

One key question our benchmark does not address is the extent to which users want diversity. For some prompts, like the dice-rolling one in Table 2, the need for diverse generations is immediately obvious. However, for other prompts, like the car recommendation example in Table 2, it may in fact be desirable for the LLM to give a consistent response each time. Should a LLM’s generations represent the opinions of a diverse group of hypothetical humans, or should they possess fixed, consistent behavior? To take the “best car to buy in 2023” example, any one human will likely repeat the same answer when asked over and over again, but across a population of humans, many different cars will be suggested.⁵ Consistency in LLM outputs is crucial for high-stakes applications like healthcare or legal assistance that rely on reliably correct answers, but it can conversely lead to the LLM imposing a single view on users.

Our findings on distributional diversity suggest that current models struggle with both objectives—they neither fully represent the diversity of valid human responses nor maintain perfect consistency across similar prompts. Future work should explore how to balance these competing goals, potentially by developing models that can explicitly modulate between diversity and consistency based on specific task requirements and user expectations. Furthermore, we would like to see user studies that investigate user interaction patterns with LLMs, how expectations of consistency/diversity vary across categories of prompts, and how users’ methods of interacting with LLM services affect perceived creativity. These findings will be useful both in guiding the behavior of future generations of language models and in developing evaluation metrics that better reflect real-world usage.

While there is starting to be some research in diversity-promoting post-training techniques, that is an area that deserves more attention from researchers. Maximum likelihood estimation inherently optimizes language models toward a single response for each prompt. This, combined with alignment algorithms which often associate a prompt with a single high-reward response, unsurprisingly causes language models to favor a certain response among several reasonable choices (Zhang et al., 2024a). Recent work has begun addressing this limitation through modified training objectives during supervised fine-tuning (Li et al., 2024) and preference optimization (Lanchantin et al., 2025), but these approaches represent only initial steps. Future work should continue exploring a broader range of methods that improve distributional diversity of language models while maintaining response quality.

⁵Human responses to the prompts in Table 2 can be found in the Appendix.

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Contributions

YZ came up with the idea of evaluating language models beyond their most likely outputs, and was responsible for running experiments.

HD, HL and VS contributed to the codebase of NOVELTYBENCH.

XL set up the website for the benchmark.

BW contributed to the collection of the NB-CURATED dataset, and was responsible for visualizations in the paper.

SH was responsible for annotating data for training and evaluating the equivalence classifier, and verified prompts in the NB-WILDCHAT subset.

DI supervised the project and suggested evaluating model creativity against humans, and proposed alternative prompting methods for inducing diversity from models.

All authors provided contributions to designing experiments, data annotation, and paper writing.

References

- Dhruv Agarwal, Mor Naaman, and Aditya Vashistha. Ai suggestions homogenize writing toward western styles and diminish cultural nuances. *arXiv preprint arXiv:2409.11360*, 2024.
- Anthropic. The Claude 3 Model Family: Opus, Sonnet, Haiku. 2024.
- Qi Cheng, Michael Boratko, Pranay Kumar Yelugam, Tim O’Gorman, Nalini Singh, Andrew McCallum, and Xiang Li. Every Answer Matters: Evaluating Commonsense with Probabilistic Measures. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 493–506, Bangkok, Thailand, August 2024. Association for Computational Linguistics.
- Jianfeng Chi, Ujjwal Karn, Hongyuan Zhan, Eric Smith, Javier Rando, Yiming Zhang, Kate Plawiak, Zacharie Delpierre Coudert, Kartikeya Upasani, and Mahesh Pasupuleti. Llama Guard 3 Vision: Safeguarding Human-AI Image Understanding Conversations, November 2024.
- Cohere. Command R and Command R+ Model Card. <https://docs.cohere.com/v2/docs/responsible-use>, October 2024.
- Flor Miriam Plaza del Arco, Amanda Cercas Curry, Alba Curry, Gavin Abercrombie, and Dirk Hovy. Angry men, sad women: Large language models reflect gendered stereotypes in emotion attribution. In *Annual Meeting of the Association for Computational Linguistics*, 2024.
- Wenchao Du and Alan W Black. Boosting Dialog Response Generation. In Anna Korhonen, David Traum, and Lluís Màrquez (eds.), *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 38–43, Florence, Italy, July 2019. Association for Computational Linguistics.
- Esin Durmus, Karina Nyugen, Thomas I Liao, Nicholas Schiefer, Amanda Askill, Anton Bakhtin, Carol Chen, Zac Hatfield-Dodds, Danny Hernandez, Nicholas Joseph, et al. Towards measuring the representation of subjective global opinions in language models. *arXiv preprint arXiv:2306.16388*, 2023.
- Ondřej Dušek, Jekaterina Novikova, and Verena Rieser. Evaluating the state-of-the-art of End-to-End Natural Language Generation: The E2E NLG challenge. *Comput. Speech Lang.*, 59(C):123–156, January 2020. ISSN 0885-2308.
- Angela Fan, Mike Lewis, and Yann Dauphin. Hierarchical Neural Story Generation. In Iryna Gurevych and Yusuke Miyao (eds.), *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 889–898, Melbourne, Australia, July 2018. Association for Computational Linguistics.
- Gemma Team et al. Gemma 2: Improving Open Language Models at a Practical Size, October 2024.
- Salvatore Giorgi, Tingting Liu, Ankit Aich, Kelsey Isman, Garrick T. Sherman, Zachary Fried, João Sedoc, Pallavi V. Kulkarni, and Brenda Curtis. Modeling human subjectivity in llms using explicit and implicit human factors in personas. In *Conference on Empirical Methods in Natural Language Processing*, 2024.
- Google. Introducing Gemini 2.0: Our new AI model for the agentic era. <https://blog.google/technology/google-deepmind/google-gemini-ai-update-december-2024/>, December 2024.
- Sil Hamilton. Detecting mode collapse in language models via narration. In Antonio Valerio Miceli-Barone, Fazl Barez, Shay Cohen, Elena Voita, Ulrich Germann, and Michal Lukasik (eds.), *Proceedings of the First edition of the Workshop on the Scaling Behavior of Large Language Models (SCALE-LLM 2024)*, pp. 65–72, St. Julian’s, Malta, March 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.scalellm-1.5/>.